Real Analysis: Answer to Exercise 4

Due date:

Question 1. **Solution**: None of the statements is true.

(a) Let $f(x) = 1/(1+x^2)$ and $g(x) = 2/(1+x^2)$. Then $f(x) < g(x)$ for all $x \in \mathbb{R}$, but $f(x) \to 0$ and $g(x) \to 0$ as $x \to \infty$; hence $a = b = 0$, so $a < b$ is false.

(b) Let $f(x) = x^3$ and $g(x) = 1/x^2$ for $x \neq 0$. Then $\lim_{x\to 0} f(x) = 0$ and $\lim_{x\to 0} g(x) = \infty$, but $f(x)g(x) = x \to 0$, not ∞.

(c) Let

$$f(x) = \begin{cases} 0, & x \neq 0, \\ 1, & x = 0. \end{cases}$$

Then the left and right limits as $x \to 0$ are both 0, but $f(0) = 1$, so $f(a) \neq \ell$.

(d) Let $g(x) = 0$ for all $x \in \mathbb{R}$ and

$$f(x) = \begin{cases} 0, & x \neq 0, \\ 1, & x = 0. \end{cases}$$

Then $\lim_{x\to 0} g(x) = 0 = b$ and $\lim_{x\to 0} f(x) = 0 = c$ (since $f(x) = 0$ for all $x \neq 0$), but $f(g(x)) = f(0) = 1$ for all x, so $\lim_{x\to 0} f(g(x)) = 1 \neq 0 = c$. Hence the statement fails.

Question 2. **Solution**: (a) As $x \to 0$, $x^2 + 1 \to 1$ and $3x - 5 \to -5$, so by the Algebra of Limits

$$\lim_{x\to 0} \frac{x^2+1}{3x-5} = \frac{1}{-5} = -\frac{1}{5}.$$

(b) Note that

$$\frac{7x}{\sin(4x)} = \frac{7}{4} \cdot \frac{4x}{\sin(4x)},$$

and as $x \to 0$, $4x \to 0$. Setting $u = 4x$,

$$\lim_{x\to 0} \frac{7x}{\sin(4x)} = \lim_{u\to 0} \frac{7}{4} \cdot \frac{u}{\sin u} = \frac{7}{4}.$$

(c) Rationalizing,

$$\frac{\sqrt{3x-2} - \sqrt{5x-6}}{\sqrt{2x-1} - \sqrt{x+1}} = -2\frac{\sqrt{2x-1} + \sqrt{x+1}}{\sqrt{3x-2} + \sqrt{5x-6}},$$

so

$$\lim_{x\to 2}\frac{\sqrt{3x-2}-\sqrt{5x-6}}{\sqrt{2x-1}-\sqrt{x+1}} = -2\cdot\frac{2\sqrt{3}}{4} = -\sqrt{3}.$$

(d) For $x \neq k\pi$,

$$\frac{x\sin x}{1-\cos x} = \frac{x(1+\cos x)}{\sin x},$$

so by the Algebra of Limits,

$$\lim_{x\to 0}\frac{x\sin x}{1-\cos x} = \left(\lim_{x\to 0}\frac{x}{\sin x}\right)\left(\lim_{x\to 0}(1+\cos x)\right) = 1\cdot 2 = 2.$$

Question 3. **Solution**: Since $|\sin(1/x^2)| \leq 1$, we have $|f(x)| \leq 2|x|$ for all x. Hence

$$|f(x)| \leq 2|x| \implies \lim_{x\to 0} f(x) = 0.$$

Question 4. **Solution**: (a) We have $-x^2 \leq x^2\cos(1/x) \leq x^2$ (since $-1 \leq \cos(1/x) \leq 1$), and $x^2 \to 0$ as $x \to 0$. By the Squeeze Theorem, $\lim_{x\to 0} x^2\cos(1/x) = 0$.

(b) Similarly $-x^4 \leq x^4\sin(1/x^2) \leq x^4$, and $x^4 \to 0$, so $\lim_{x\to 0} x^4\sin(1/x^2) = 0$.

Question 5. **Solution**: (a) Using $\sin u/u \to 1$ as $u \to 0$,

$$\frac{\sin 3x}{\sin 5x} = \frac{\sin 3x}{3x}\cdot\frac{5x}{\sin 5x}\cdot\frac{3}{5} \to 1\cdot 1\cdot\frac{3}{5} = \frac{3}{5}.$$

(b) Using $\tan u/u \to 1$ as $u \to 0$,

$$\frac{\tan 2x}{x} = 2\cdot\frac{\tan 2x}{2x} \to 2.$$

Question 6. **Solution**: (a) For $x > 0$, as $x \to 0^+$, $1/x$ becomes arbitrarily large and positive, so $\lim_{x\to 0^+} 1/x = +\infty$. For $x < 0$, as $x \to 0^-$, $1/x$ becomes arbitrarily large in magnitude and negative, so $\lim_{x\to 0^-} 1/x = -\infty$. Since the one-sided limits are not equal (in fact they diverge in opposite directions), $\lim_{x\to 0} 1/x$ does not exist.

(b) Dividing numerator and denominator by x^2,

$$\lim_{x\to\infty}\frac{2x^2+3x-1}{x^2-4} = \lim_{x\to\infty}\frac{2+3/x-1/x^2}{1-4/x^2} = \frac{2}{1} = 2.$$